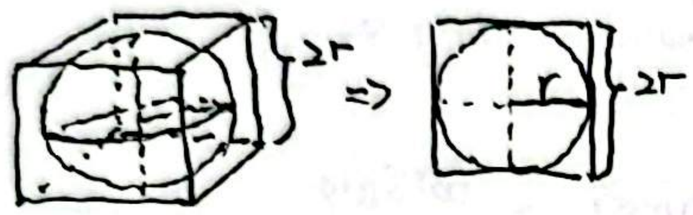
Lecture : CS548T 07 – Dimensionality & Linear Dimensionality Reduction.

The quality of calculated BDR depends on the CCD estimates.

High dimensional spaces are weird! Do Not trust your intuition.

- Consinder a hypersphere (an inscribed) a hypercube :



Volume sphere : $V_d(r) = \frac{\pi^{d/2} r^d}{\Gamma(\frac{d}{2}+1)}$

Volume hypercube : $V_d = (2r)^d$

where $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$

$\Gamma(n+1) = n! \quad (n>0)$

let $f_d = \frac{V_d(r)}{(2r)^d} = \frac{\pi^{d/2}}{2^d \Gamma(\frac{d}{2}+1)}$

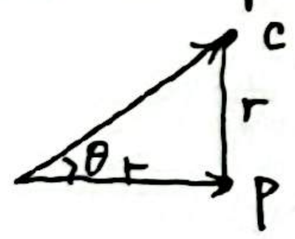
d	1	2	3	....	∞
f_d	1	0.79	0.52	....	0

as $n \to \infty$, $f_d \to 0$

- Consinder the cornur

$c = [r, r, \dots r]^T$

and axis $p = [c, 0, \dots 0]^T$

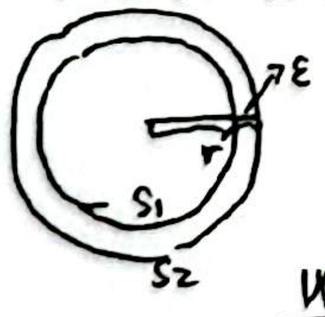


$\cos\theta = \frac{c^T p}{\|c\| \cdot \|p\|} = \frac{r^2}{r\sqrt{d} \cdot r} = \frac{1}{\sqrt{d}}$

$d \to \infty$, $\cos\theta \to 0$

$\Rightarrow$ axis $\perp$ cornur ($\theta \to \frac{\pi}{2}$)

- Consinder a hypersphere :



$V_{shell} = V(S_2) - V(S_1)$

$= \left(1 - \frac{V(S_1)}{V(S_2)}\right) V(S_2)$

$\frac{V(S_1)}{V(S_2)} = (1-\frac{\varepsilon}{r})^d$

$0 < 1 - \frac{\varepsilon}{r} < 1$

$\Rightarrow d \to \infty \quad \frac{V(S_1)}{V(S_2)} \to 0.$

$V_{shell} \to V(S_2)$?

- High-dimensional Gaussian :

Let $X \sim N(0, \sigma^2 I_d)$ i.e. $x_i \sim N(0, \sigma^2)$ iid.

$E[\|x\|^2] = \sigma^2 d$

$E[\frac{1}{d}\|x\|^2] = \sigma^2$

$\frac{1}{d}\|x\|^2 \sim N(\sigma^2, \frac{1}{d})$

Curse of Dimensionality (维度灾难) :

In theory, adding new features will not increase p(error).

But in practice, for BDR, error increases as new fatures are added (since in high dimensions, CCD estimates get worse).

desired training set size $O(e^p)$, p = # parameters.

高维 => #parameter 多 => 估计难 => need more data.

Solutions :

① Reduce # of parameters (complexity of model)

② Reduce # of features (dimensionality reduction)
→ implicitly reduce # parameter.

③ create more data

(a) Bayesian formulation.

(b) data augmentation :

Linear dimensionality Reduction :

- summarize correlated features with fewer features.

① Principal Component Analysis : (PCA).

idea : if the data lives in a subspace, it will look flat in some directions, if we fit a Gaussian, it will be highly skewed (skinny ellipse).

Consinder the matrix $\Sigma = V \Lambda V^T$.

$V = [v_1, v_2 \dots v_d]$ eigenvectors

$\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{bmatrix}$ eigenvalues

for (v_i, λ_i)

each v_i define an axis of ellipse

each λ_i define the width of ellipse axis.

Eigenvalues tells us which directions are flat :

Select v_i with largest eigenvalues

PCA: Given dataset $\{x_1, x_2 \cdots x_n\}$ (dim k) $\to$ to

(1) Calculate Gaussian:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i \quad \Sigma = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T$$

(2) eigen decomp: $\Sigma = V \Lambda V^T$

(3) sor eigenvalue: $\lambda_1 > \cdots > \lambda_d > 0$

(4) select top k eigenvectors: $\Phi = [v_1 \cdots v_k]$

(5) project x onto Φ: $z = \Phi^T(x-\mu) \in \mathbb{R}^k$
(PCA coefficients)

(6) z as new feature vector $\to$ BDR as visual.

Notes: This selection of Φ with $\Phi^T\Phi = I$

(1) maximizes the variance of projected training data. $\left(\sum_i^n \|z_i\|^2\right)$

(2) minimizes the reconstruction error

$$\sum \|x_i - \hat{x}_i\|^2$$

$$\hat{x}_i = \Phi(z_i + \mu)$$

- can be implemented efficiently using SVD.

(1) pick a k that works

(2) pick k to preserve $p\%$ of variance of data $p = \frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{n}\lambda_i}$

- Assumption: "noise" variance is smaller than the "signal" variance

PCA is optimal for representation, but not necessarily classification.

No way to fix it (don't use class information).

Linear Discriminant Analysis: (LDA)

Fisher Linear Discriminant (FLD)

find linear project. that best separates the classes. $z = w^T x$

Class mean: $\mu_j = \frac{1}{n_j}\sum_{x_i \in C_j} x_i$

Class scatter: $S_j = \sum_{x_i \in C_j}(x_i-\mu_j)(x_i-\mu_j)^T$

1-d(z):

$m_j = w^T\mu_j$

$s_j = w^T S_j w$.

IDEA: $(m_1-m_2)^2 = (w^T(\mu_1-\mu_2))^2$

(w is unconstrained).

Fisher idea

$$w^* = \arg\max_w \frac{(m_1-m_2)^2}{s_1+s_2}$$

$$\vdots$$

$$= \arg\max_w \frac{w^T S_B w}{w^T S_W w}$$

$S_B = (\mu_1-\mu_2)(\mu_1-\mu_2)^T$

$S_W = S_1 + S_2$

$w^* = S_W^{-1}(\mu_1-\mu_2) = (S_1+S_2)^{-1}(\mu_1-\mu_2)$

this hyperplane that separates 2 Gaussians with cov S_1+S_2

FLD is optimal when 2 classes are Gaussian with equal cov.